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Reviewed work(s):
Source: *Journal of the American Statistical Association*, Vol. 43, No. 244 (Dec., 1948), pp. 572-574
Published by: American Statistical Association
Stable URL: http://www.jstor.org/stable/2280710
Accessed: 08/03/2013 09:36

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A TEST FOR SYMMETRY IN CONTINGENCY TABLES

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SUPPOSE THAT a sample of N individuals is classified according to two arguments, A and B, in an m by m contingency table and suppose that the categories of classification A are associated in some unique way with the categories of classification B. For example, in the study in which this problem arose the categories were related to the position of plates of stickleback (Gasterosteus) and the two arguments corresponded to right and left sides; it was logical to associate identical patterns on opposite sides. Denote by P_{ij} the probability that a randomly chosen individual belongs to the ith row and jth column of the table. As usual, a dot indicates summation over the replaced index.

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	Category	Classification A 1	2	—	—	—	m	Marginal Distribution
Classification B	1	P_{11}	P_{12}	—	—	—	P_{1m}	$P_{1.}$
	2	P_{21}	P_{22}				P_{2m}	$P_{2.}$
	—	—	—				—	
	—	—	—				—	
	—	—	—				—	
	—	—	—				—	
	m	P_{m1}	P_{m2}				P_{mm}	$P_{m.}$
Marginal Distribution		$P_{.1}$	$P_{.2}$				$P_{.m}$	

It is desired to test the hypothesis that the above table is symmetric (i.e., $P_{ij}=P_{ji}$). If this hypothesis is true, then $P_{k.}=P_{.k}$; that is, the marginal distribution of individuals is the same for classifications A and B. Further, the conditional distribution with respect to A for the kth category of B is equal to the conditional distribution with respect to B for the kth category of A. This equality of conditional distributions in general implies a stronger equivalence between the classifications than equality of marginal distributions, although for $m=2$ equality of marginal distributions implies complete symmetry. The weaker hypothesis of equality of marginal distributions would also be of interest, especially in the absence of symmetry; this problem appears to be somewhat more difficult to handle by straightforward methods.

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The proposed test for the hypothesis of symmetry is of the familiar χ^2 type. Let n_{ij} be the observed frequency in the ith row and jth column of a table of observed quantities arranged as above. Then

$$(1) \qquad \sum_{i>j}\sum \frac{(n_{ij} - n_{ji})^2}{n_{ij} + n_{ji}}$$

has for large N the Chi-square distribution with $m(m-1)/2$ degrees of freedom. A large value of χ^2 leads to rejection of the hypothesis. Note that the value of χ^2 is computed from non-diagonal frequencies, which is reasonable since diagonal frequencies should not afford evidence tending to disprove the hypothesis of symmetry.

In cases in which the observed frequencies are small and no regrouping of frequencies is possible (e.g., in a 2×2 table) an exact test may be of interest. Specifically, in a 2×2 table an exact test may be made by testing with any table of the binomial distribution the hypothesis that n_{12} and n_{21} are observed "successes" and "failures" in a random sample of $n_{12}+n_{21}$ events. From equation (1) we obtain for a 2×2 table the following χ^2 statistic with one degree of freedom:

$$(2) \qquad \frac{(n_{12} - n_{21})^2}{n_{12} + n_{21}}.$$

This is equivalent to the value obtained by a test of goodness of fit of a binomial with $P=\frac{1}{2}$. Formula (2) was obtained by McNemar [1], who lists applications in psychological testing. A problem in the same field prompted a recent query in *The American Statistician* [2].

Several intuitive derivations of the above test may be given, but a rigorous justification may be obtained from a theorem of Cramér [3], page 426, which puts the use of χ^2 tests in contingency tables on a sound theoretical basis. This procedure affords an interesting example of a routine application of the general theory and might well be an exercise for a course in the theory of χ^2 tests. After verifying certain regularity conditions, the method of Cramér can be applied by computing the quantity

$$(3) \qquad \sum_i \sum_j \frac{(n_{ij} - NP_{ij})^2}{NP_{ij}}$$

where the P_{ij} are replaced by their maximum likelihood estimates under the hypothesis of symmetry. These estimates turn out to be $n_{ij}+n_{ji}/2N$ which yield equation (1) on substitution into (3).

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An alternate derivation is available if (1) is regarded as a sum of $m\ (m-1)/2$ terms of type (2). This approach requires the assumption that we are sampling from a population with fixed sums for all symmetric pairs of cells. While both methods lead to the same result, the more general one based on Cramér's theorem seems slightly preferable. Until the test procedure is derived, it is not obvious that the use of the conditional distribution with fixed totals for symmetric pairs entails no loss of efficiency.

The problem which prompted this note was suggested by Professor Frank Weymouth.

REFERENCES

[1] McNemar, Quinn, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, Vol. 12, No. 2, 1947, pp. 153–157.

[2] "Question 1," *American Statistician*, Vol. 1, No. 2, 1947, p. 12.

[3] Cramér, Harald, *Mathematical Methods of Statistics*, Princeton University press, 1946.

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